

Neurons and neuroglia

The written notes for the neurons and neuroglia walkthrough. Week 4 competencies 7 to 11.

To print: these notes as a two-column PDF.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

The map of the nervous system

- The central nervous system (CNS) is the brain and spinal cord. It is where information is integrated and decisions are made.
- The peripheral nervous system (PNS) is everything else: the nerves that carry information in and commands out.
- The sensory division, also called the afferent division, carries information toward the CNS from receptors in the skin, muscles, eyes, ears and internal organs. Most signals from the organs never reach awareness.
- The efferent division carries commands away from the CNS to effectors, which are muscles and glands.

Efferent branch	What it controls	Example
Somatic motor division	Skeletal muscle	Picking up a pencil on purpose; many reflexes
Autonomic division (visceral nervous system)	Smooth muscle, cardiac muscle, glands, some adipose tissue	Heart rate speeding up without your deciding it

- The autonomic division has two branches that often have opposite effects on the same organ. The organ's activity reflects the balance between them.
- Sympathetic branch: fight-or-flight. Prepares the body for stress and effort, for example raising heart rate and sending more blood to working muscles.
- Parasympathetic branch: rest-and-digest. Runs maintenance such as digestion, and slows the heart.
- Enteric nervous system: a network of neurons in the wall of the digestive tract. It can sense, decide and command on its own, and the autonomic division normally adjusts it.

Afferent

Carrying toward the CNS. The sensory division.

Efferent

Carrying away from the CNS, toward effectors.

Effector

A muscle or gland that carries out a command.

Neurons sorted by direction and by shape

By direction	What it does	Where
Sensory (afferent) neuron	Carries information toward the CNS	Receptor to CNS
Interneuron	Connects other neurons; does the processing, comparing and deciding	Entirely inside the CNS
Efferent (motor) neuron	Carries commands away from the CNS	CNS to effector

- Every neuron has dendrites, short branches that receive signals, and one axon, a long fiber that carries the signal away.
- Neurons are sorted by shape by counting the processes that leave the cell body.

Shape	Processes	Where it is found
Multipolar	Many dendrites and one axon	Interneurons of the brain and spinal cord, and efferent neurons such as somatic motor neurons (cell bodies in the ventral horn)
Bipolar	One dendrite and one axon	Retina and olfactory epithelium
Pseudounipolar	One stalk that splits in two	Sensory neurons for touch, pain and stretch; cell bodies in the dorsal root ganglia

- In a pseudounipolar neuron the signal runs from the skin straight past the cell body toward the spinal cord, without detouring through the cell body where it could fade.
- Sensory fibers enter the spinal cord through the dorsal root and motor axons leave through the ventral root, so damage to one root affects one direction only.

The four working regions of a neuron

1. Dendrites receive signals from other cells. One neuron may receive input from thousands of others. The signals here are graded potentials: small, local, variable in size, and they fade as they spread.
2. The cell body (soma) holds the nucleus and makes the cell's proteins. Signals from every branch spread into it and add together, which is called summation.
3. The trigger zone, the axon hillock and the initial segment of the axon, decides whether to fire. It has the highest density of voltage-gated sodium channels in the neuron, so it reaches threshold first. In a sensory neuron the trigger zone sits near the receptor endings instead.
4. The axon carries action potentials. An action potential is all or none: it is rebuilt at full size as it travels, so it arrives as big as it started. A motor axon from the spinal cord to the foot can be about a meter long.
5. The axon terminals release neurotransmitter. When an action potential arrives, voltage-gated calcium channels open,

calcium rushes in, and the rise in calcium makes vesicles fuse with the membrane and release neurotransmitter.

- A stronger stimulus does not make a bigger action potential. It makes action potentials fire more often, and it recruits more neurons.
- If the calcium channels at the terminal are blocked, the action potential arrives normally but no neurotransmitter is released, so the message stops at the terminal.

Graded potential

A small, local change in membrane voltage whose size varies with the input and which fades with distance.

Action potential

A large, all-or-none signal that travels the length of the axon without getting smaller.

Trigger zone

The axon hillock and initial segment, where the neuron decides whether to fire.

Synapse

The junction between an axon terminal and the next cell: a neuron, a muscle fiber or a gland.

The glial cells

Glial cell	Where	Job
Astrocyte	CNS	Most common CNS glia. Controls the fluid around neurons: takes up extra potassium after neurons fire, clears leftover neurotransmitter such as glutamate, passes fuel from blood to neurons. Its end feet signal capillary cells to form the tight junctions of the blood-brain barrier.
Oligodendrocyte	CNS	Makes myelin. One oligodendrocyte wraps segments of several axons.
Microglia	CNS	Resident immune cells. Patrol, then activate to engulf invaders, damaged cells and debris.
Ependymal cell	CNS	Lines the ventricles and central canal. Specialized ependymal cells make cerebrospinal fluid; cilia help move it.
Schwann cell	PNS	Makes myelin. One Schwann cell wraps one segment of one axon. Helps cut peripheral axons regrow.
Satellite cell	PNS	Forms a capsule around cell bodies in ganglia and regulates the fluid around them. A kind of Schwann cell that does not make myelin.

- Blood-brain barrier: brain capillaries are sealed by tight junctions. Small lipid-soluble molecules (oxygen, carbon dioxide, alcohol) cross easily; water-soluble molecules (glucose, amino acids) need transport proteins. It keeps out many toxins, and many drugs.
- If astrocytes cannot clear potassium, outside potassium rises, the resting potential becomes less negative, and neurons sit closer to threshold. This is one link to seizure disorders.
- Hydrocephalus: if cerebrospinal fluid cannot drain, it builds up, the ventricles swell and pressure rises on the brain. Fluid stays steady only when production matches drainage.
- Because one oligodendrocyte myelinates many axons, losing one damages insulation on many fibers.

When an axon is cut

1. The part of the axon cut off from the cell body dies, because it is no longer supplied.
2. In the PNS, Schwann cells survive, clear debris, line up into a tube and release growth factors, and the axon can regrow along that path, at roughly a millimeter a day.
3. In the CNS, oligodendrocytes and astrocytes release molecules that block regrowth, and astrocytes form a scar.

Myelin and what happens when it is lost

- Myelin is a glial cell's own membrane wound around the axon many times. Membrane is mostly lipid and ions cannot cross it, so myelin is electrical insulation.
- Myelin covers the axon in segments. The bare gaps between them are the nodes of Ranvier.
- Voltage-gated sodium channels are packed at the nodes, with very few in the membrane under the myelin.
- The slow step in conduction is the membrane opening channels to rebuild the signal, not current moving along the inside of the axon. Myelin stops current leaking out, so current from one node reaches the next node still strong enough to rebuild the signal there.
- The signal is rebuilt only at the nodes, so it effectively jumps from node to node. This is saltatory conduction.
- In electrical terms, myelin raises membrane resistance (less current leaks out) and lowers membrane capacitance (less charge is needed to change the voltage). A wider axon also conducts faster, because it has less resistance to current flowing along the inside.

Axon	Conduction speed
Large, myelinated	Up to about 120 m/s
Thin, unmyelinated	About 0.5 to 2 m/s

Demyelination

1. The myelin over a stretch of axon is destroyed and the bare axon is exposed.
2. Current leaks out of the bare stretch.
3. That membrane was always covered, so it has very few sodium channels to rebuild the signal.
4. If the stretch is long enough, the signal fades before the next node reaches threshold. The message never arrives: conduction block.

Disease	Myelin attacked	Pattern
Multiple sclerosis	CNS myelin, made by oligodendrocytes	Scattered, unrelated-looking symptoms (vision, sensation, strength, balance) that come and go as patches become inflamed and partly recover. Warmth can briefly tip barely conducting fibers into block.
Guillain-Barré syndrome	PNS myelin, made by Schwann cells	Weakness that climbs the legs, often after an infection; reflexes lost; conduction slowed but response size near normal. Most people recover over weeks to months because Schwann cells remyelinate well.

Axonal transport

- Proteins are made in the cell body. Nothing is built at the axon terminals, so everything has to be shipped along the axon.

- Cargo travels on microtubules that run the length of the axon. Motor proteins walk along them, splitting ATP at every step. If the microtubules are taken apart, traffic stops in both directions.

Direction	Motor protein	Cargo
Anterograde: cell body to terminals	Kinesin	Synaptic vesicles, new membrane proteins such as channels, mitochondria
Retrograde: terminals to cell body	Dynein	Worn-out membrane and organelles for recycling, growth factor signals from the target cell

Speed	Rate	Carries
Fast transport	Up to about 400 mm per day	Vesicles and mitochondria
Slow transport	About 0.2 to 2.5 mm per day	Cytoskeleton proteins and dissolved enzymes

- At 400 mm per day, fast transport covers a 1 meter axon in about 2.5 days.
- A cut peripheral axon regrows about a millimeter a day because it is built from cytoskeleton proteins that arrive by slow transport.
- A block on the axon piles outward cargo up on the cell body side. Blocking the inward motor piles returning cargo up on the terminal side.
- Retrograde transport is also a way in: rabies virus and tetanus toxin ride it from the terminals into the CNS. Herpes viruses ride it in to hide in a sensory cell body, and later ride anterograde transport back out, so a cold sore returns in the same spot.